VisionQuant

Satellite-Based Alternative Data for Freight Equity Prediction

Moonkyu Jun, Seunghoon Jang

FINANCE TRACK

AI Community DATATHON

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Investors Are Flying Blind — We Watch From Space

The Problem

- Traditional investors rely on **lagging indicators** — earnings reports arrive after the fact
- Quarterly filings are already **priced in** by the time they're public
- No real-time window into **operational activity**

Our Solution

- Satellite imagery of **Memphis FedEx hub** and **Louisville UPS hub**
- **YOLOv11** detects aircraft and trucks on the tarmac
- Physical cargo activity converted into **quantifiable financial signals**

From Pixels to Profit

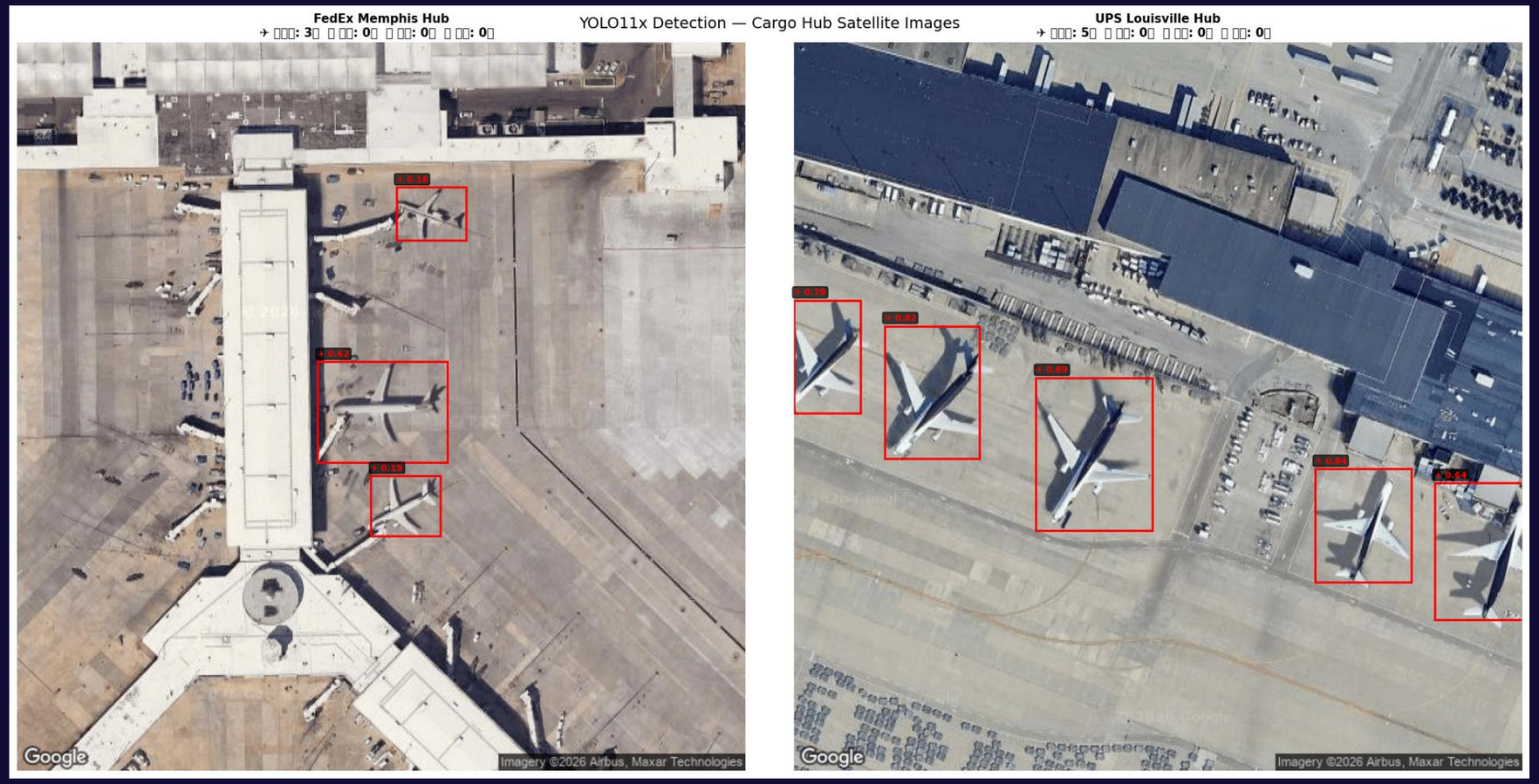
Hubs Analyzed

- Memphis, TN — FedEx World Hub (FDX)
- Louisville, KY — UPS Worldport (UPS)

Data Sources

- Google Earth Engine, NAIP 1m/pixel
- Sentinel-2 multispectral imagery
- yfinance for equity price validation

From Pixels to Profit



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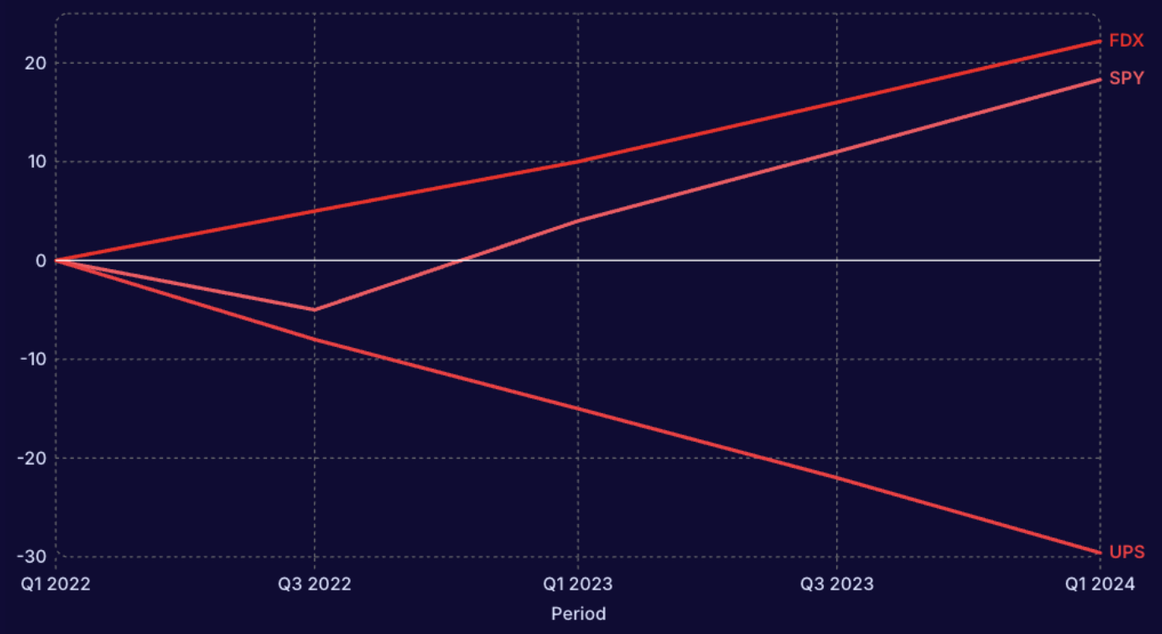
Same Sector, Opposite Directions

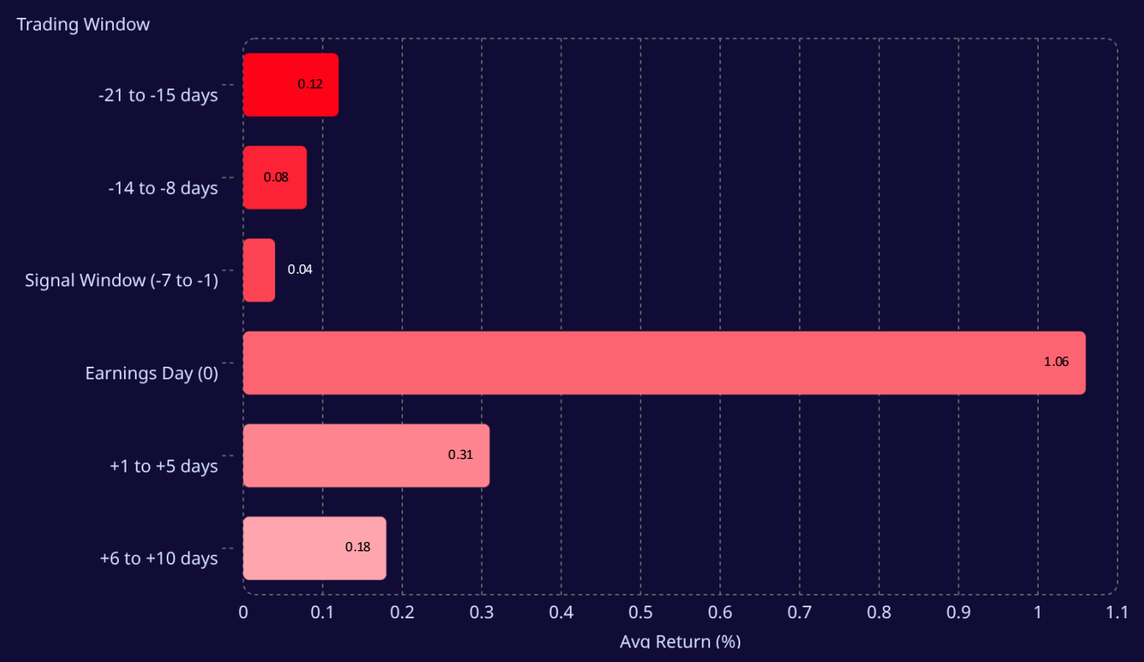


The Divergence

Two freight giants in the same sector — moving in **opposite directions** over 2.5 years. Can satellite cargo activity explain this divergence?

FDX	UPS	SPY
+22.2%	-29.6%	+18.3%

The Market Reacts Late — We Don't

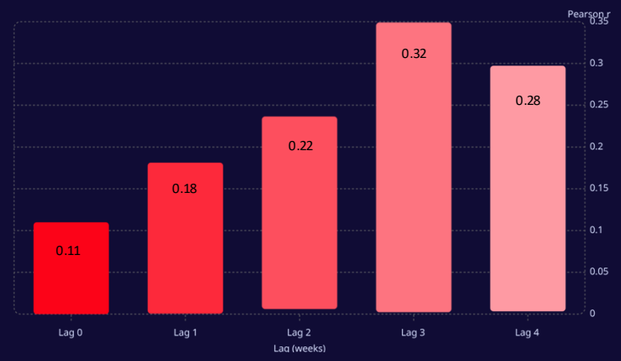


Key Insight

During the **3–4 week satellite signal window** before earnings, market returns are near zero — the market has **no information yet.**

Earnings Day Avg Return: +1.06%
The market wakes up — but our signal fires 3 weeks earlier.

Satellite Activity Leads Stock Price by 3 Weeks



Statistically Validated

r = 0.387–0.443

Peak correlation at 3 week lag

p < 0.001

Statistically significant signal

n = 130+

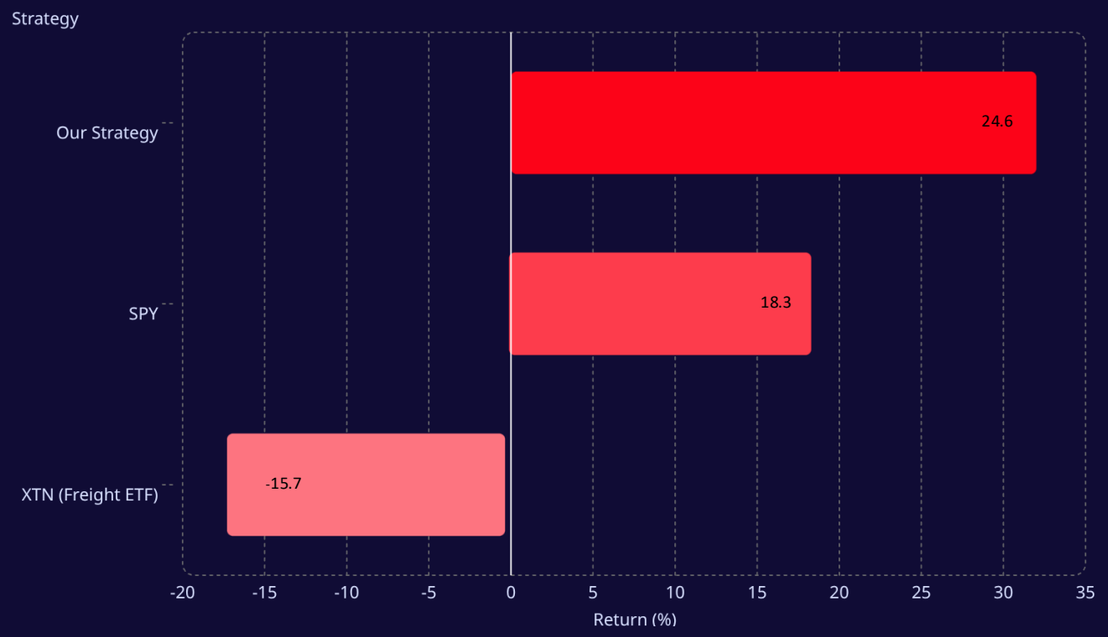
Data points across 2.5 years

Predicting Direction. Beating the Market.

Signal Strategy

- YOLOv11 object detection on NAIP 1m/pixel imagery
- Composite activity index derived from satellite signals
- BUY signal: activity index > 0.35
- SELL signal: activity index < 0.15
- Applied to FDX (Memphis) and UPS (Louisville)

Backtest Performance



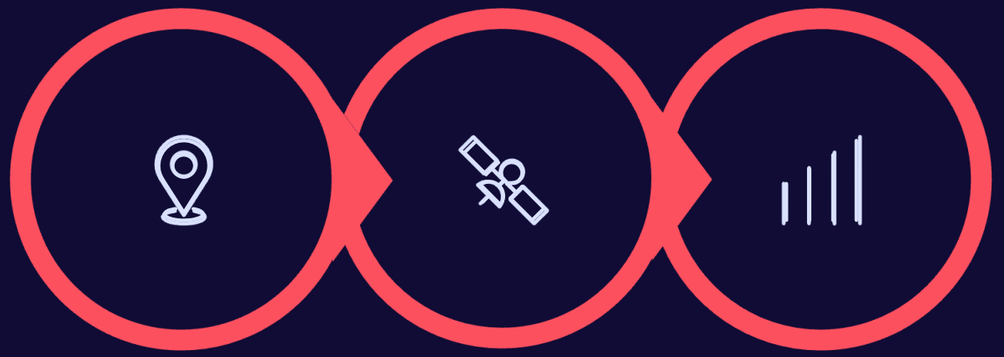
FDX Strategy

+24.6% return. Outperformed SPY by +6.3%p

UPS Correlation

r = 0.443, p < 0.001. Lag 1 week — statistically significant

30 Seconds to a Signal



Click Hub

Satellite + YOLOv11

BUY / SELL / HOLD

"Anyone can use it. No Bloomberg terminal required."
Deployable from a browser, accessible to any analyst or portfolio manager.

Limitations

Current Limitations

- **10m resolution** limits precise vehicle-level detection accuracy
- **Cloud cover** creates intermittent data gaps in time series
- **Correlation ≠ Causation** — confounders require rigorous control
- **2.5 years** of data limits generalization across market regimes

Future Roadmap

01

Planet Labs 3m/pixel

Near-daily high-resolution imagery

02

Real-Time Pipeline

Live signal generation with automated ingestion

03

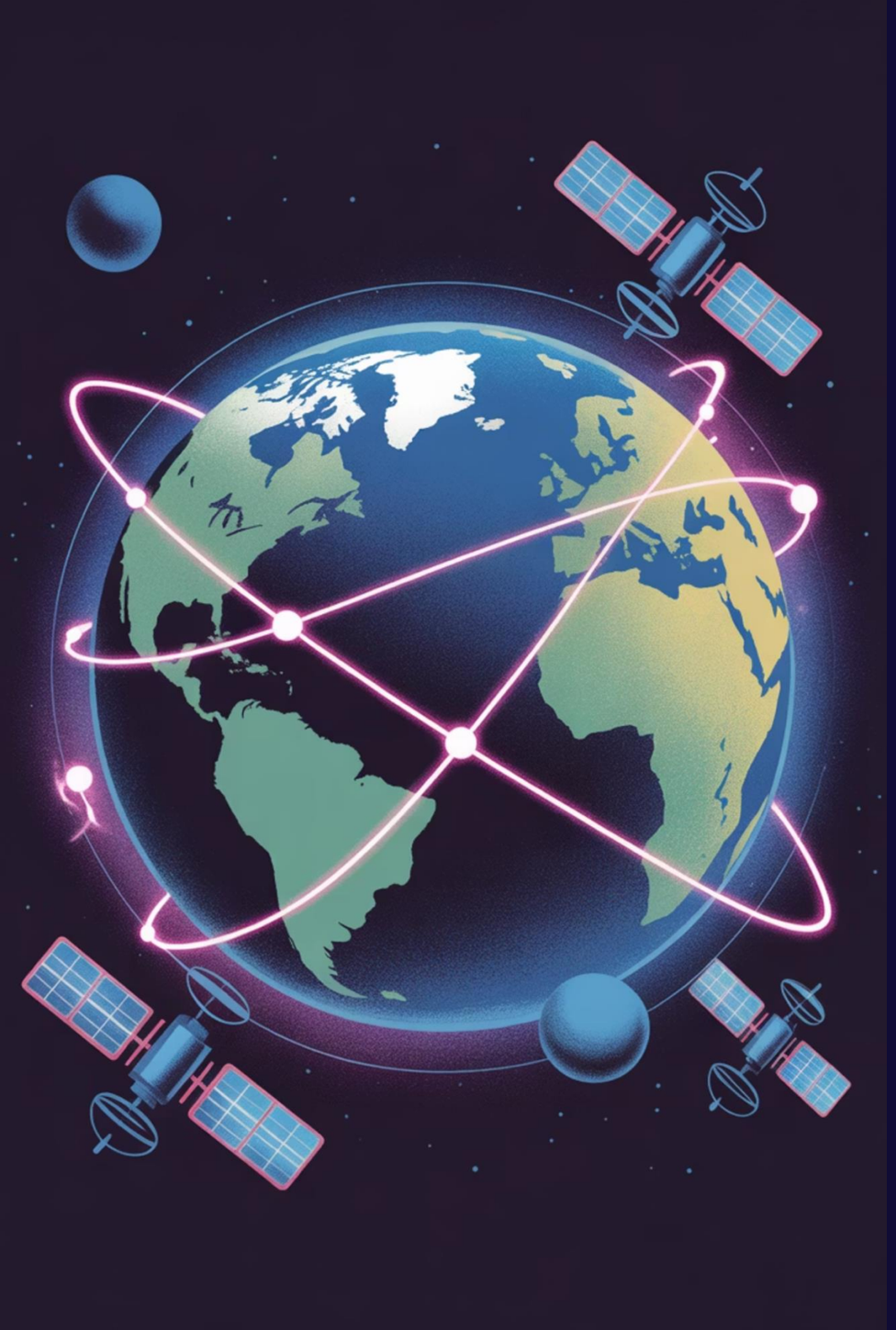
Multi-Factor Model

Satellite + fuel prices + FX rates combined

04

Global Expansion

International hubs: Dubai, Singapore, Amsterdam



We See What the Market Doesn't

Predictive Lead Time

Satellite cargo activity leads freight equity prices by **3–4** weeks

Statistical Rigor

Validated at **$p < 0.001$**, Pearson **$r = 0.443$**, $n = 130+$ observations

Model Edge

FDX satellite strategy: +24.6% return vs SPY +18.3% (+6.3%p outperformance)

> "The edge is in the data nobody else is looking at."

VISIONQUANT TEAM